

## CHURANOV, Czech Republic

WMO: 114570

Lat: 49.0682N

Lon: 13.6152E

Elev: 3682

StdP: 12.84

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 5.9        | 10.2 | -1.2                          | 5.9 | 14.8 | 4.0 | 7.8 | 17.3 | 26.7                  | 31.9 | 24.2 | 30.6 | 5.9                   | 80   | 0.630 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.1                   | 77.5            | 61.2 | 74.1 | 60.3 | 70.9 | 59.2 | 63.3                | 73.0 | 61.8 | 71.1 | 60.3 | 68.9 | 6.3                  | 60   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 59.9                            | 88.5 | 66.7 | 58.4 | 83.7 | 64.9 | 57.0 | 79.4 | 63.3 | 30.7          | 73.1 | 29.5 | 71.1 | 28.4 | 69.3 | 71.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 21.0              | 18.4 | 16.1 | 1.4                        | 83.2 | 4.9                | 3.1 | -2.1                                               | 85.4 | -4.9       | 87.3 | -7.6       | 89.0 | -11.2      | 91.3 |
|                   |      |      | 0.9                        | 66.8 | 4.8                | 1.9 | -2.5                                               | 68.1 | -5.3       | 69.3 | -8.0       | 70.3 | -11.5      | 71.7 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 42.1   | 26.4  | 27.5  | 32.5  | 40.9  | 48.8  | 54.8  | 58.1  | 57.9  | 50.0  | 43.3  | 35.2  | 28.7 | (6)  |
| (7)  |                                                                                  | DBStd       | 13.86  | 8.31  | 9.11  | 8.22  | 8.63  | 7.46  | 7.39  | 6.70  | 6.68  | 6.90  | 8.27  | 8.70  | 8.31 | (7)  |
| (8)  |                                                                                  | HDD50       | 3795   | 731   | 631   | 542   | 292   | 115   | 35    | 9     | 10    | 85    | 239   | 447   | 659  | (8)  |
| (9)  |                                                                                  | HDD65       | 8403   | 1196  | 1051  | 1006  | 724   | 504   | 315   | 230   | 237   | 450   | 673   | 893   | 1124 | (9)  |
| (10) |                                                                                  | CDD50       | 910    | 0     | 0     | 1     | 19    | 76    | 178   | 261   | 254   | 86    | 31    | 4     | 0    | (10) |
| (11) |                                                                                  | CDD65       | 43     | 0     | 0     | 0     | 0     | 1     | 8     | 17    | 16    | 1     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH74       | 305    | 0     | 0     | 0     | 0     | 5     | 64    | 120   | 109   | 7     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH80       | 35     | 0     | 0     | 0     | 0     | 0     | 3     | 16    | 16    | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 7.5    | 9.3   | 8.8   | 8.2   | 6.7   | 6.6   | 6.4   | 6.7   | 6.2   | 6.5   | 7.4   | 8.1   | 9.4  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 35.2   | 2.6   | 2.3   | 2.8   | 1.9   | 3.1   | 4.0   | 4.2   | 3.8   | 2.7   | 2.5   | 2.5   | 2.8  | (15) |
| (16) |                                                                                  | PrecMax     | 50.8   | 5.9   | 5.4   | 7.0   | 4.2   | 5.3   | 7.2   | 6.7   | 11.6  | 5.9   | 6.9   | 5.1   | 7.7  | (16) |
| (17) |                                                                                  | PrecMin     | 25.8   | 0.2   | 0.3   | 0.6   | 0.2   | 0.6   | 1.3   | 1.1   | 0.9   | 0.8   | 0.7   | 0.1   | 0.9  | (17) |
| (18) |                                                                                  | PrecStd     | 5.4    | 1.5   | 1.4   | 1.7   | 0.8   | 1.3   | 1.4   | 1.5   | 2.0   | 1.3   | 1.5   | 1.3   | 1.5  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 48.0  | 51.6  | 58.0  | 68.3  | 73.7  | 79.8  | 82.0  | 82.4  | 73.7  | 67.6  | 59.6  | 50.2 | (19) |
| (20) |                                                                                  |             | MCWB   | 37.8  | 37.7  | 44.1  | 51.3  | 57.8  | 61.7  | 62.4  | 61.3  | 59.0  | 53.5  | 47.8  | 40.7 | (20) |
| (21) |                                                                                  | 2%          | DB     | 43.4  | 46.1  | 52.1  | 62.0  | 68.9  | 75.9  | 77.2  | 76.8  | 68.0  | 62.1  | 54.6  | 45.0 | (21) |
| (22) |                                                                                  |             | MCWB   | 37.0  | 37.4  | 41.9  | 49.6  | 56.6  | 61.1  | 61.6  | 61.1  | 57.6  | 51.8  | 44.8  | 38.0 | (22) |
| (23) |                                                                                  | 5%          | DB     | 40.2  | 42.1  | 47.7  | 58.0  | 65.3  | 71.6  | 73.5  | 72.8  | 64.5  | 58.2  | 50.3  | 41.7 | (23) |
| (24) |                                                                                  |             | MCWB   | 36.5  | 36.1  | 40.0  | 47.6  | 54.7  | 59.6  | 60.4  | 60.2  | 56.1  | 50.4  | 43.7  | 37.4 | (24) |
| (25) |                                                                                  | 10%         | DB     | 37.3  | 38.8  | 43.6  | 54.0  | 61.5  | 67.9  | 69.9  | 69.5  | 60.8  | 54.4  | 46.7  | 38.9 | (25) |
| (26) |                                                                                  |             | MCWB   | 34.6  | 34.9  | 38.6  | 45.4  | 52.5  | 57.6  | 59.3  | 59.2  | 54.4  | 48.9  | 42.2  | 36.1 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 41.6  | 41.9  | 45.6  | 53.5  | 61.0  | 65.7  | 65.2  | 64.6  | 61.4  | 56.0  | 49.6  | 42.7 | (27) |
| (28) |                                                                                  |             | MCD B  | 43.1  | 47.3  | 53.5  | 64.0  | 69.3  | 74.5  | 74.9  | 74.6  | 68.9  | 62.6  | 55.4  | 46.0 | (28) |
| (29) |                                                                                  | 2%          | WB     | 39.1  | 39.3  | 43.3  | 50.5  | 58.1  | 62.6  | 63.2  | 62.8  | 58.7  | 53.7  | 46.8  | 40.4 | (29) |
| (30) |                                                                                  |             | MCD B  | 41.4  | 42.6  | 49.3  | 60.2  | 66.7  | 71.5  | 73.3  | 72.7  | 65.8  | 59.2  | 52.0  | 43.1 | (30) |
| (31) |                                                                                  | 5%          | WB     | 36.9  | 37.3  | 41.3  | 48.4  | 55.6  | 60.5  | 61.6  | 61.2  | 56.8  | 51.6  | 44.7  | 38.6 | (31) |
| (32) |                                                                                  |             | MCD B  | 39.4  | 40.9  | 46.0  | 56.2  | 63.1  | 69.5  | 70.9  | 70.7  | 62.8  | 56.2  | 48.9  | 40.9 | (32) |
| (33) |                                                                                  | 10%         | WB     | 34.6  | 35.4  | 39.3  | 46.4  | 53.4  | 58.5  | 60.1  | 59.7  | 55.0  | 49.6  | 42.7  | 36.5 | (33) |
| (34) |                                                                                  |             | MCD B  | 36.9  | 38.6  | 42.8  | 53.3  | 60.1  | 66.5  | 68.2  | 68.2  | 59.8  | 53.8  | 46.5  | 38.5 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.2   | 10.4  | 11.6  | 14.3  | 14.8  | 15.3  | 15.1  | 12.8  | 12.0  | 10.1  | 9.0   | (35) |      |
| (36) |                                                                                  |             | MCD BR | 13.3  | 15.3  | 18.1  | 19.2  | 19.9  | 20.6  | 20.2  | 20.2  | 18.0  | 17.2  | 16.1  | 13.4 | (36) |
| (37) |                                                                                  |             | MCWBR  | 8.8   | 9.6   | 10.7  | 10.5  | 10.9  | 10.8  | 9.4   | 9.5   | 9.9   | 10.3  | 9.5   | 9.0  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 10.8  | 12.8  | 15.4  | 17.8  | 18.2  | 18.6  | 17.8  | 18.3  | 15.9  | 15.4  | 14.5  | 11.3 | (38) |
| (39) |                                                                                  |             | MCWBR  | 8.2   | 9.0   | 10.0  | 10.4  | 10.8  | 10.5  | 9.3   | 9.3   | 9.4   | 9.9   | 9.4   | 8.8  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.265  | 0.285 | 0.319 | 0.350 | 0.355 | 0.357 | 0.365 | 0.359 | 0.336 | 0.313 | 0.289 | 0.264 | (40) |      |
| (41) |                                                                                  | taud        | 2.421  | 2.407 | 2.394 | 2.364 | 2.375 | 2.388 | 2.360 | 2.394 | 2.439 | 2.499 | 2.517 | 2.475 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 258    | 278   | 284   | 285   | 287   | 286   | 282   | 279   | 275   | 264   | 247   | 244   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 20     | 26    | 31    | 36    | 36    | 36    | 37    | 34    | 30    | 23    | 18    | 16    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 296    | 582   | 912   | 1352  | 1579  | 1735  | 1664  | 1467  | 1046  | 640   | 315   | 226   | (44) |      |
| (45) |                                                                                  | RadStd      | 52     | 81    | 120   | 173   | 186   | 149   | 163   | 150   | 140   | 75    | 36    | 28    | (45) |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |      |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |      |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |      |
| (46) | Station Only            | N/A     | +2.05  | N/A     | N/A   | N/A   | +0.61       | N/A   | N/A   | N/A   | N/A | (46) |
| (47) | Regional (82 neighbors) | +0.83   | N/A    | N/A     | N/A   | +0.67 | +0.63       | N/A   | N/A   | +108  | +47 | (47) |

Nomenclature: See separate page